

Customer Churn Prediction – Data Science Case Study

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1. Problem statement

A subscription-based company wants to reduce customer churn by predicting which customers are at risk and understanding the drivers behind churn.

2. Dataset

Telco customer churn dataset with customer demographics, contract information, charges, and a binary churn flag (Yes/No).

3. Methodology (high level)

- Performed exploratory data analysis (EDA) to understand churn distribution and key relationships (tenure, charges, contract type).
- Cleaned and preprocessed data, encoded categorical variables, and standardized numeric variables.
- Trained two supervised learning models: Logistic Regression and Random Forest.
- Evaluated models using accuracy, precision, recall, F1-score, and ROC AUC on a held-out test set.
- Selected the best model based on ROC AUC and inspected feature importances to extract insights.

4. Best model performance

- Best model: Logistic Regression
- Accuracy: 0.804
- Precision: 0.648
- Recall: 0.572
- F1: 0.608
- Roc_auc: 0.836

5. Feature importance: best model is Logistic Regression; inspect coefficients in the notebook for details.

6. Business recommendations (summary)

- Use the best model to score customers regularly and focus interventions on those with the highest predicted churn risk.
- Pay particular attention to high-value customers (for example high monthly charges) with high churn scores.
- For short-tenure customers and those on flexible contracts, review onboarding, communications, and value proposition.
- Consider model and data improvements such as hyperparameter tuning and adding behavioural features.

End of report.